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Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection

MAJOR PROJECT REPORT (CS481P)

Submitted by,

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Under the guidance of

Dr. Nagaraja G.S

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Professor

Assistant Professor

Dept. of CSE

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RV College of Engineering

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In partial fulfillment for the award of degree of

Bachelor of Engineering

in

Computer Science and Engineering

Department of CSE

2025-26





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CERTIFICATE

Certified that the major project (CS481P) work titled ***Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection*** is carried out by **Nikhil Vasu (1RV22CS128)**, **Rohith Biradar (1RV22CS164)** and **Suraj Chanaveeragoudra (1RV22CS211)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

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DECLARATION

We, **Nikhil Vasu, Rohith Biradar** and **Suraj Chanaveeragoudra** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Federated Heterogeneous Graph Neural Networks for Trade-Based Money Laundering Detection**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

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ABSTRACT

Trade-Based Money Laundering (TBML) has emerged as a significant challenge within modern financial systems due to the increasing complexity and scale of global trade transactions. Conventional Anti-Money Laundering (AML) systems primarily rely on isolated rule-based monitoring frameworks that are often ineffective in identifying hidden transactional dependencies and multi-hop laundering typologies across interconnected banking networks. To address these limitations, this work developed a privacy-preserving Federated Heterogeneous Graph Neural Network (HGNN)-based framework for real-time multi-class TBML detection across distributed institutional environments.

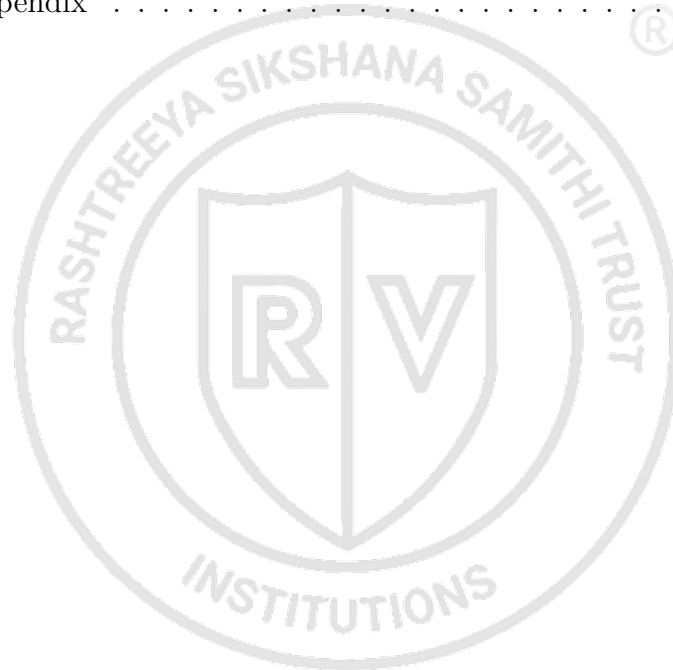
The proposed framework modeled accounts, trade entities, SWIFT transactions, and trade documents as heterogeneous financial graphs using Memgraph. A hybrid analytical architecture integrating Graph Attention Network (GAT)-based convolutional layers and Long Short-Term Memory (LSTM)-based temporal sequence modeling was employed to capture relational topology and evolving transactional behavior. Federated learning using the Federated Averaging (FedAvg) aggregation strategy enabled collaborative cross-bank model optimization without centralized transactional data sharing. The implemented workflow incorporated Apache Kafka-based transaction streaming, graph construction, federated fraud inference, Suspicious Transaction Report (STR) generation, and regulator-facing compliance synchronization using Python, PyTorch Geometric, FastAPI, PostgreSQL, Memgraph, and React.js.

Experimental evaluation across distributed banking clients demonstrated stable federated convergence with Fraud-class Precision values approaching 0.98, F1-Scores nearing 0.84, and Precision-Recall Area Under Curve (PR-AUC) values exceeding 0.92. Concurrent scalability evaluation additionally maintained dashboard retrieval latency near 156–162 ms and fraud-filtering response latency near 148–159 ms under increasing institutional workloads, thereby validating the operational suitability of the proposed framework for scalable and privacy-preserving TBML detection. The framework can be extended into a fully integrated banking compliance platform with automated risk response, transaction control, and regulatory workflows. These enhancements would transform it from a monitoring system into a complete financial crime prevention ecosystem.

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ABBREVIATIONS

AI Artificial Intelligence

AML Anti-Money Laundering

ERC Electrical Rule Check

FedAvg Federated Averaging

FL Federated Learning

GAT Graph Attention Network

GCN Graph Convolutional Network

GNN Graph Neural Network

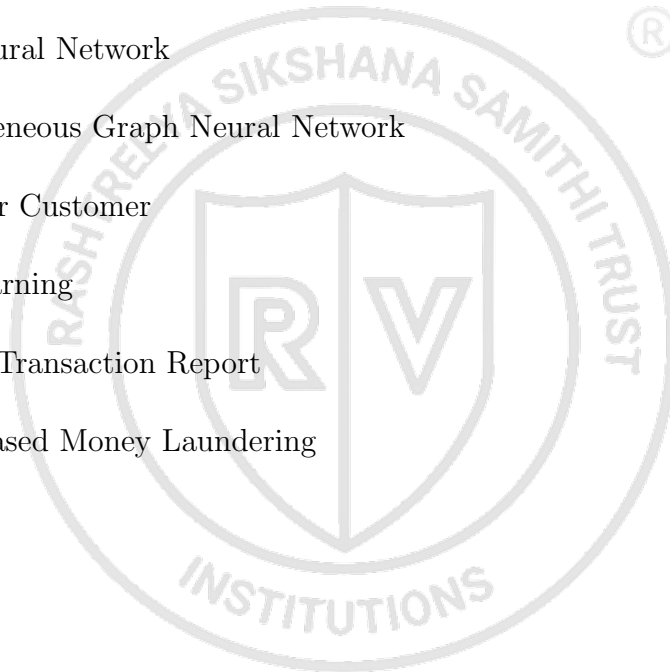
HGNN Heterogeneous Graph Neural Network

KYC Know Your Customer

ML Machine Learning

STR Suspicious Transaction Report

TBML Trade-Based Money Laundering



LIST OF SYMBOLS

λ Wavelength of a light in a medium

τ Time constant

c Speed of light

f Frequency of light in a medium





Chapter 1

Software Requirement Specification

CHAPTER 1

SOFTWARE REQUIREMENT SPECIFICATION

The development of an intelligent Trade-Based Money Laundering detection framework requires careful identification of the operational, computational, and functional requirements associated with the system. Since the proposed framework integrates graph-based fraud detection, federated learning, real-time monitoring, and compliance management mechanisms, defining the system requirements becomes essential for ensuring scalability, security, and efficient analytical performance. This chapter presents the Software Requirement Specification (SRS) of the proposed TBML detection framework, including the overall system description, product functionalities, user characteristics, functional and non-functional requirements, and the software and hardware resources required for implementing the proposed system.

1.1 Overall Description

The proposed system is designed as a privacy-preserving financial monitoring platform for detecting suspicious Trade-Based Money Laundering activities within large-scale transactional ecosystems. The framework combines heterogeneous graph-based fraud analysis, federated learning mechanisms, role-based monitoring dashboards, and automated Suspicious Transaction Report (STR) generation within a unified analytical environment. The system supports real-time transaction monitoring, fraud classification, and interactive visualization of suspicious financial activities to assist compliance analysts and financial institutions in identifying hidden laundering patterns efficiently.

1.1.1 Product Perspective

The proposed TBML detection framework functions as an intelligent financial monitoring and fraud analysis platform capable of integrating machine learning-based anomaly detection with interactive compliance monitoring systems. The architecture combines heterogeneous graph representation, Graph Neural Network-based analytical models, federated learning mechanisms, and modern web-based visualization technologies within a unified system environment.

The system represents financial entities such as accounts, organizations, invoices, and transactions as interconnected graph structures for relational analysis. Fraud detection modules analyze these relationships to identify suspicious transactional dependencies

and hidden laundering patterns that are difficult to detect using conventional rule-based Anti-Money Laundering systems. The framework additionally integrates role-based dashboards for transaction monitoring, fraud visualization, alert management, and automated compliance reporting within financial institutions.

1.1.2 Product Functions

The primary functionalities supported by the proposed framework are summarized as follows:

- **Transaction Monitoring:** The system continuously monitors financial transactions and trade-related activities to identify suspicious transactional behavior within large-scale financial ecosystems.
- **Graph-Based Financial Analysis:** Financial entities such as accounts, organizations, invoices, and transactions are represented as interconnected graph structures for relational analysis and hidden dependency detection.
- **Fraud Classification and Risk Prediction:** Graph Neural Network-based analytical models classify suspicious transactions and generate fraud risk predictions based on transactional relationships and behavioral patterns.
- **Federated Learning Support:** The framework supports collaborative model training across multiple institutional environments while preserving sensitive financial data privacy.
- **Real-Time Dashboard Visualization:** Interactive monitoring dashboards provide visualization of transaction analytics, fraud alerts, risk distributions, and suspicious activity summaries for compliance analysts.
- **Suspicious Transaction Report (STR) Generation:** The system automatically generates structured Suspicious Transaction Reports for identified fraudulent transactions to support compliance and regulatory workflows.
- **Authentication and Role-Based Access Control:** Secure authentication and controlled access mechanisms are incorporated to manage user privileges and protect sensitive financial information within the monitoring platform.

1.1.3 User Characteristics

The proposed TBML detection framework supports multiple categories of institutional users involved in fraud monitoring, compliance verification, and regulatory supervision activities. Since the system processes sensitive transactional and compliance-related information, role-based access mechanisms are enforced to ensure secure and controlled interaction with the platform. The major user categories associated with the proposed framework are summarized as follows:

- **Bank Compliance Team:** The bank compliance team represents institutional analysts responsible for monitoring transactional activities and investigating suspicious financial behavior identified by the fraud-detection framework. These users are expected to possess domain knowledge related to Anti-Money Laundering (AML) procedures, transaction analysis, and institutional compliance operations. Authorized compliance personnel interact with analytical dashboards, review suspicious transaction records, verify fraud alerts, and manage institutional STR escalation workflows.
- **Regulatory Administrator:** The regulatory administrator functions as a supervisory authority responsible for overseeing fraud-monitoring activities across participating financial institutions. These users typically possess advanced regulatory and compliance expertise associated with financial investigation and institutional auditing operations. Regulatory administrators access consolidated Suspicious Transaction Reports (STRs), monitor institution-level compliance activities, and supervise large-scale suspicious transactional behavior across distributed banking environments.

1.1.4 Constraints

The development and deployment of the proposed TBML detection framework are subject to multiple operational, computational, and regulatory constraints associated with financial monitoring systems. These limitations influence the scalability, analytical performance, and real-time operational capabilities of the framework.

The major constraints associated with the proposed system are summarized as follows:

- **Data Privacy and Regulatory Constraints:** Financial institutions operate

under strict privacy regulations and compliance policies that restrict direct sharing of sensitive transactional information across organizations.

- **Large-Scale Transactional Data Processing:** The system must process large volumes of interconnected financial transactions and graph relationships, which may increase computational complexity and memory utilization during real-time analysis.
- **Availability of Real-World TBML Datasets:** Access to publicly available Trade-Based Money Laundering datasets is limited due to confidentiality and regulatory restrictions, making large-scale real-world experimentation challenging.
- **Dynamic Fraud Patterns:** Money laundering strategies continuously evolve over time, requiring fraud detection models to adapt to changing transactional behaviors and emerging laundering techniques.
- **Infrastructure and Computational Limitations:** Training graph-based machine learning models and processing heterogeneous financial networks require significant computational resources, especially for large-scale transactional ecosystems.

1.2 Specific Requirements

The specific requirements of the proposed TBML detection framework define the operational functionalities, performance expectations, software dependencies, and computational resources required for efficient execution of the system. These requirements ensure that the framework can support real-time fraud analysis, scalable transaction monitoring, secure user interaction, and intelligent compliance management within financial institutions.

1.2.1 Functional Requirements

The functional requirements define the major operations and services that must be supported by the proposed TBML detection framework. The primary functional requirements of the system are summarized as follows:

- The system shall enforce role-based authorization policies to regulate operational access privileges for compliance analysts and regulatory administrators.

- The framework shall support ingestion and preprocessing of transactional datasets for heterogeneous financial graph construction.
- The proposed system shall model financial entities, transactions, and trade relationships as interconnected heterogeneous graph structures.
- The analytical engine shall perform graph-based fraud classification and generate transaction-level risk predictions using federated learning methodologies.
- The dashboard infrastructure shall support real-time visualization of suspicious transactions, fraud alerts, analytical summaries, and compliance-monitoring activities.
- The system shall enable institutional compliance teams to review suspicious transactions and perform fraud escalation or clearance procedures.
- The framework shall provide features to generate Suspicious Transaction Reports (STRs) for transactions identified as potentially fraudulent.
- The regulator-side monitoring interface shall support centralized supervision of institutional fraud reports and suspicious financial activities.
- The analytical interface shall provide transaction-filtering, search, and investigative querying functionalities for fraud-analysis workflows.

1.2.2 Non-Functional Requirements

The non-functional requirements define the quality attributes, operational constraints, and performance expectations associated with the proposed TBML detection framework. These requirements ensure that the system remains scalable, secure, reliable, and responsive while processing large-scale financial transactions and analytical workloads.

The primary non-functional requirements associated with the proposed system are summarized as follows:

- **Scalability:** The framework should support scalable processing of large transactional datasets and heterogeneous graph structures without significant degradation in analytical performance.

- **Real-Time Processing:** The system should support low-latency transaction analysis and fraud prediction to enable continuous real-time monitoring of suspicious financial activities.
- **Security and Privacy:** The framework should ensure secure access control, protected communication, and privacy-preserving handling of sensitive financial information and compliance-related data.
- **Reliability:** The system should maintain stable analytical performance and continuous monitoring capabilities during prolonged operational workloads and concurrent user access.
- **Availability:** The monitoring platform should provide consistent access to dashboards, fraud analytics, and reporting services for authorized institutional users.
- **Maintainability:** The system architecture should support modular development, efficient debugging, and simplified integration of additional analytical modules and monitoring functionalities.
- **Usability:** The dashboard interfaces and visualization systems should provide intuitive interaction mechanisms for compliance analysts and regulatory administrators during fraud investigation activities.

1.2.3 Software Requirements

The implementation of the proposed TBML detection framework requires multiple software technologies, machine learning libraries, backend services, database systems, and visualization frameworks to support graph-based fraud detection, federated learning, real-time transaction monitoring, and compliance management functionalities. The proposed framework integrates both analytical and monitoring components that require coordinated interaction between frontend dashboards, backend processing services, database systems, and machine learning environments. These technologies collectively support scalable fraud analysis, secure transaction management, and efficient visualization of suspicious financial activities. The software requirements associated with the proposed system are summarized in Table 1.1.

Table 1.1: Software Requirements of the Proposed TBML Detection Framework

Category	Technologies / Tools	Purpose in the System
Operating System	Windows 11	Provides the development and deployment environment for backend services, databases, and analytical modules.
Programming Languages	Python, JavaScript	Used for backend development, machine learning workflows, API services, and frontend application development.
Frontend Frameworks	Next.js, React.js, Tailwind CSS	Supports responsive user interface development, dashboard visualization, and real-time monitoring functionalities.
Backend Framework	FastAPI	Facilitates asynchronous API communication between frontend dashboards and backend fraud detection services.
Database Systems	PostgreSQL, Memgraph	Stores transactional records, graph relationships, compliance information, and analytical data required for fraud analysis.
Machine Learning Frameworks	PyTorch, PyTorch Geometric	Supports deep learning operations, Graph Neural Network implementation, and graph-driven fraud analysis.
Federated Learning Framework	Flower	Enables distributed federated learning and privacy-preserving model aggregation across institutional environments.
Streaming Platform	Apache Kafka	Supports real-time transaction streaming and event-driven processing within the monitoring pipeline.
Containerization Platform	Docker	Provides containerized deployment and scalable execution environments for system integration.
LLM Integration Service	Groq LLM API	Supports automated generation of Suspicious Transaction Reports and compliance-oriented analytical summaries.

1.2.4 Hardware Requirements

The implementation of the proposed TBML detection framework requires the following hardware components to support graph processing, machine learning model execution, federated learning coordination, real-time transaction monitoring, and analytical dashboard visualization. The hardware requirements associated with the proposed system are summarized in Table 1.2.

Table 1.2: Hardware Requirements of the Proposed TBML Detection Framework

Hardware Component	Minimum Requirement	Purpose in the System
Processor	Intel Core i7 / AMD Ryzen 7	Supports efficient execution of graph analytical models, backend services, and machine learning computations.
RAM	16 GB	Supports graph datasets, real-time analytical workloads, and concurrent monitoring operations.
GPU	NVIDIA RTX 3050	Accelerates Graph Neural Network training and deep learning computations.
Storage	512 GB SSD	Stores transactional datasets, graph databases, analytical models, and generated compliance reports.

1.3 Summary

In this chapter, the Software Requirement Specification of the proposed Trade-Based Money Laundering detection framework was presented. The operational functionalities, user roles, functional and non-functional requirements associated with the proposed fraud detection platform were discussed to establish the monitoring and analytical capabilities of the system.

The chapter also identified the software technologies, database systems, machine learning frameworks, and hardware resources required for implementing the proposed framework. These specifications collectively establish the operational and computational foundation required for developing the proposed TBML detection system, which is further discussed in the subsequent system design and implementation chapter.



Chapter 2

High Level Design

CHAPTER 2

HIGH LEVEL DESIGN

The design of the proposed Trade-Based Money Laundering detection framework focuses on developing a scalable, privacy-preserving, and intelligent financial monitoring system capable of analyzing complex transactional relationships within large-scale trade ecosystems. Since modern TBML activities involve interconnected financial entities, cross-border transactions, and evolving laundering strategies, the proposed framework integrates graph-based analytical models, federated learning mechanisms, and real-time monitoring services to support efficient fraud detection and compliance management. This chapter presents the major design considerations, architectural strategies, overall system architecture, and data flow mechanisms associated with the proposed fraud detection framework.

2.1 Design Considerations

The design of the proposed TBML detection framework is influenced by multiple operational, analytical, and security-related factors associated with large-scale financial monitoring systems. Since Trade-Based Money Laundering activities involve interconnected financial entities, cross-border transactions, and hidden transactional dependencies, the system architecture must support scalable analytical processing, secure institutional collaboration, and efficient graph-based fraud analysis. The major design considerations associated with the proposed framework are discussed in the following subsections.

2.1.1 General Considerations

The proposed TBML detection framework is designed to support large-scale transaction monitoring and fraud analysis across distributed financial environments. Since laundering activities typically involve interconnected accounts, trade entities, invoices, and transactional relationships, the framework utilizes graph-based modeling techniques to represent hidden financial dependencies more effectively than conventional relational systems. The architecture additionally supports real-time transaction processing and scalable analytical workflows for continuously increasing institutional transaction volumes.

Security, privacy preservation, and system scalability are important considerations within the proposed architecture due to the sensitive nature of financial and compliance-

related information. To address these requirements, the framework incorporates secure authentication mechanisms, role-based access control, and federated learning strategies for protected inter-institutional collaboration without centralized transactional data sharing. The overall system is developed using a modular architecture to simplify integration, maintenance, and future enhancement of transaction monitoring, fraud detection, compliance reporting, and dashboard visualization services.

2.1.2 Development Methods

The proposed TBML detection framework is developed using a modular and iterative development methodology to support scalable system integration and continuous refinement of analytical functionalities. Individual system components including transaction monitoring services, graph construction pipelines, fraud detection models, federated learning environments, and dashboard interfaces are developed and tested independently before integration within the complete monitoring framework.

The development process additionally incorporates incremental model evaluation and experimental refinement to improve fraud detection performance and analytical reliability. Graph-based machine learning models are trained using processed transactional datasets, while frontend and backend services are integrated through API-based communication mechanisms to support real-time interaction between monitoring modules and analytical services.

Version control and collaborative development practices are utilized throughout the implementation process to manage backend services, machine learning workflows, frontend interfaces, and analytical configurations efficiently. This modular development methodology simplifies debugging, testing, maintenance, and future enhancement of the proposed fraud detection framework.

2.2 Architectural Strategies

The architectural strategies adopted in the proposed TBML detection framework focus on supporting scalable financial transaction analysis, secure institutional collaboration, and efficient interaction between analytical services and monitoring interfaces. The framework integrates graph-based fraud detection models, backend analytical services, real-time dashboards, and distributed learning mechanisms within a unified monitoring environment. The major architectural strategies associated with the proposed framework

are discussed in the following subsections.

2.2.1 Programming Language

The proposed TBML detection framework primarily utilizes Python for backend services, graph processing, machine learning workflows, and analytical computations. Python was selected due to its extensive support for deep learning libraries, graph analytical frameworks, and federated learning environments. Libraries such as PyTorch, PyTorch Geometric, and Flower are utilized for implementing graph neural network models and distributed learning mechanisms within the fraud detection framework.

The frontend monitoring dashboard is developed using Next.js and React.js to support responsive user interface development and real-time analytical visualization. JavaScript and TypeScript are utilized for implementing dashboard components, transaction monitoring interfaces, and compliance management services. FastAPI is additionally integrated to support API-based communication between frontend dashboards, backend services, and machine learning modules.

2.2.2 System Integration Architecture

The proposed framework follows a modular client-server and service-oriented integration architecture to support coordinated interaction between analytical modules, backend services, and frontend monitoring interfaces. The fraud detection models, transaction monitoring services, authentication modules, and compliance reporting components operate as independent services while maintaining synchronized communication through backend APIs and shared database environments.

The machine learning modules responsible for graph processing and fraud prediction operate independently from the frontend monitoring dashboard. Processed fraud predictions, transaction classifications, and generated Suspicious Transaction Reports are synchronized through backend services to support real-time visualization and compliance management within the monitoring environment. This modular integration strategy simplifies scalability, maintenance, and future extension of the proposed framework.

2.2.3 Data Storage Management

The proposed TBML detection framework utilizes both relational and graph-based database systems to support efficient financial data management and analytical processing. PostgreSQL is utilized for storing structured transactional records, authentication

information, compliance-related data, and generated Suspicious Transaction Reports. The relational database environment supports secure storage and efficient retrieval of operational monitoring information within the framework.

Memgraph is utilized as the graph database environment for storing interconnected financial entities and transactional relationships. Accounts, organizations, invoices, and transactions are represented as graph nodes and edges to support relational fraud analysis and graph neural network processing. The combined relational and graph-based storage strategy improves scalability, analytical flexibility, and efficient synchronization between backend analytical services and monitoring dashboards.

2.3 Overall System Architecture

The proposed TBML detection framework is designed to enable intelligent financial transaction monitoring, graph-based fraud analysis, federated learning, and automated compliance reporting across distributed institutional environments. It combines transaction processing, heterogeneous graph construction, machine learning-based fraud detection, and real-time monitoring to identify suspicious activities related to Trade-Based Money Laundering.

The architecture begins with multiple data sources, including banking systems, trade platforms, and external financial providers. Transaction records, trade documents, KYC information, and monitoring logs are collected and processed through a transaction processing module that performs data validation, cleansing, normalization, and event generation for analytical use.

The processed data is then converted into heterogeneous graph structures, where entities such as accounts, organizations, invoices, and transactions are represented as interconnected nodes and edges. This graph representation helps capture hidden financial relationships and suspicious transaction patterns required for graph-based learning.

The fraud detection layer consists of a Hybrid Graph Neural Network module for relational analysis and a temporal learning module for sequential transaction analysis. Together, these components learn complex laundering behaviors and improve fraud detection accuracy through combined risk prediction.

To ensure privacy and regulatory compliance, the framework adopts a federated learning approach where institutions train models locally without sharing sensitive financial data. Only model updates are transmitted to a global aggregation module, enabling

collaborative learning while preserving data confidentiality.

The aggregated global model is used within the inference and risk scoring module to classify suspicious transactions and generate fraud risk scores. Transactions identified as high-risk are forwarded to the compliance management system for alert generation, Suspicious Transaction Report creation, and regulatory support. The complete architecture of the proposed TBML detection framework is illustrated in Figure 2.1.

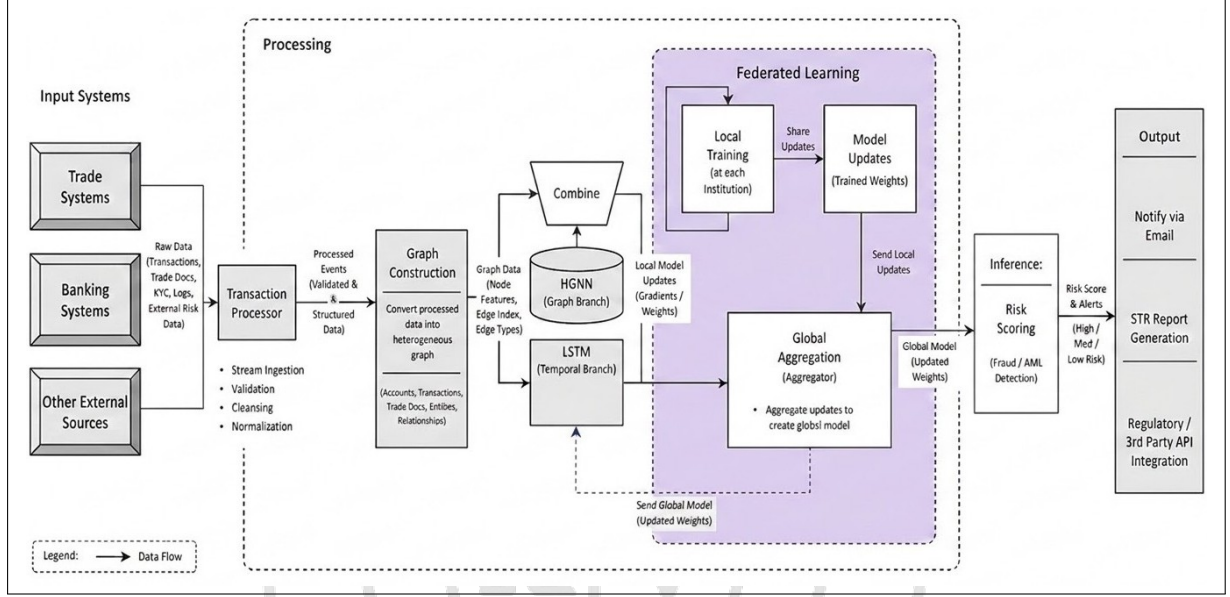


Figure 2.1: Overall Architecture of the Proposed TBML Detection Framework

2.4 Data Flow Diagrams

A Data Flow Diagram (DFD) is a graphical representation used to illustrate the movement of data within a system and the interactions between processes, external entities, and data repositories. DFDs provide a structured visualization of how information enters, is processed, stored, and transferred throughout a system architecture. In the proposed Federated Heterogeneous Graph Neural Network (HGNN)-Based Trade-Based Money Laundering (TBML) Detection System, DFDs are utilized to model the overall workflow of transaction acquisition, graph construction, federated learning, fraud analysis, and alert generation. The hierarchical decomposition of the system into Level 0, Level 1, and Level 2 diagrams enables a detailed understanding of the operational flow, subsystem interactions, and data processing mechanisms involved in real-time AML analytics. The diagrams presented in this section are designed using the Yourdon and De Marco DFD methodology to ensure standardized representation and structured system

decomposition.

2.4.1 Data Flow Diagram Level 0

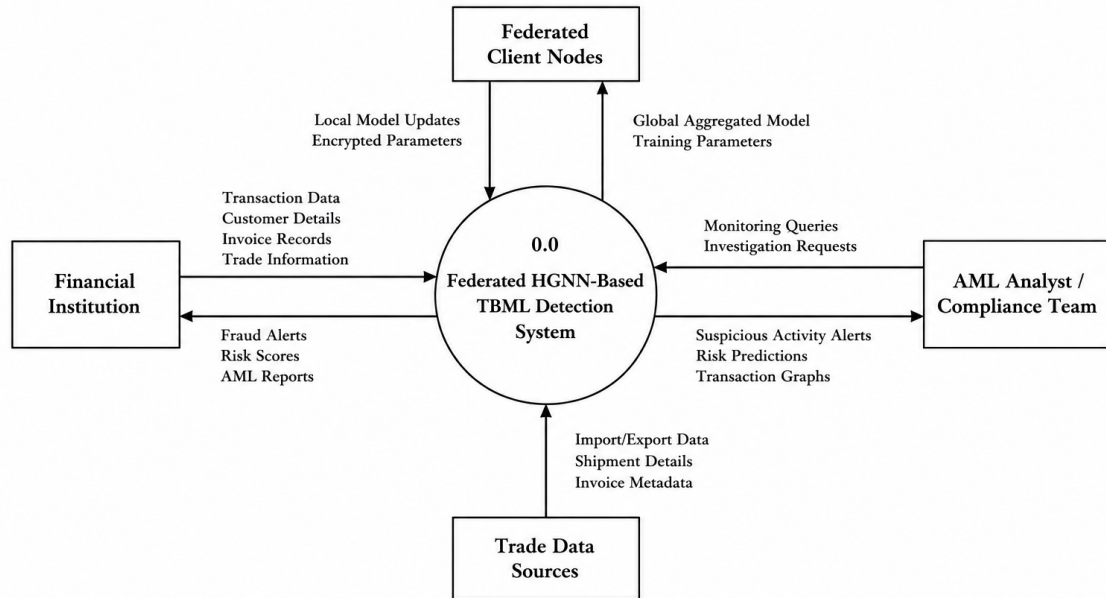


Figure 2.2: Level 0 Data Flow Diagram

Figure 4.2 illustrates the Level 0 Data Flow Diagram of the proposed Federated HGNN-Based TBML Detection System. The diagram provides a high-level contextual representation of the complete system as a single process interacting with external entities including Financial Institutions, Federated Client Nodes, AML Analysts, and Trade Data Sources. Financial institutions provide transaction records, customer information, invoice metadata, and trade-related details to the system for analysis. Trade data sources contribute import-export records and shipment information required for transaction validation and graph construction. Federated client nodes exchange encrypted local model updates and receive aggregated global model parameters during distributed training. The system processes these inputs to generate fraud alerts, risk predictions, AML reports, and suspicious activity notifications for compliance teams and financial analysts. The Level 0 DFD establishes the overall system boundary and illustrates the primary flow of information involved in distributed TBML detection and real-time financial fraud analytics.

2.4.2 Data Flow Diagram Level 1

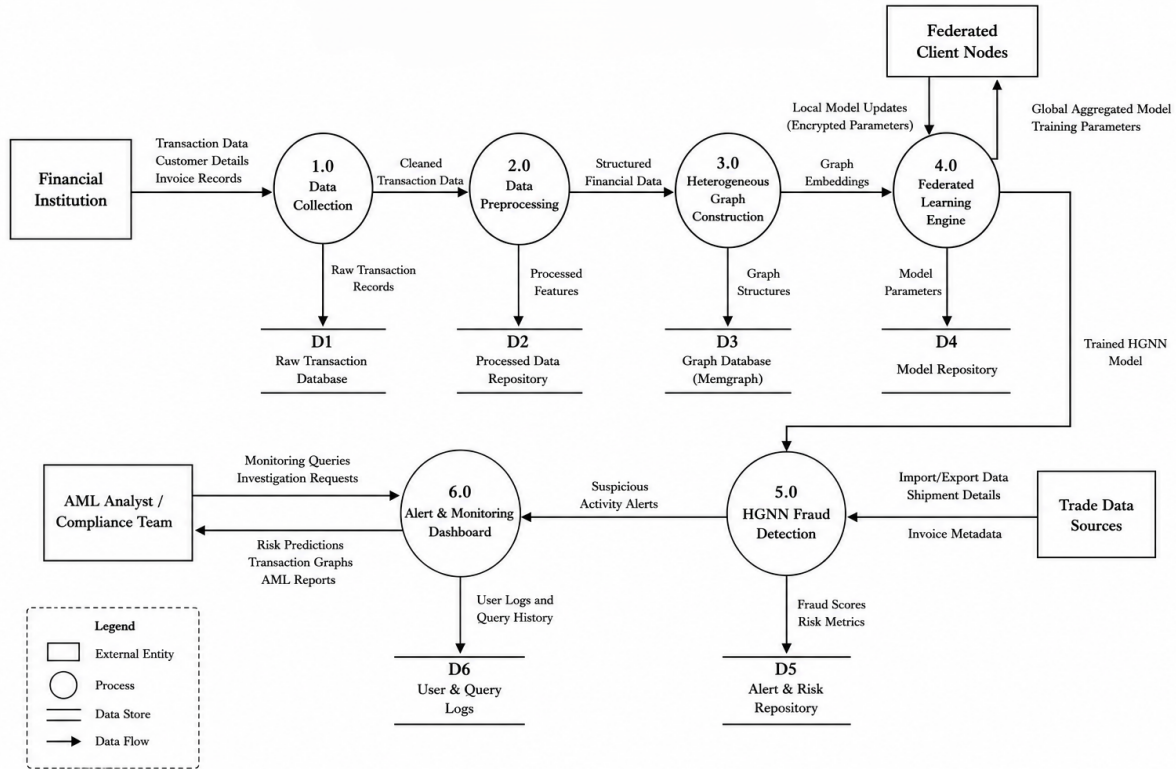


Figure 2.3: Level 1 Data Flow Diagram

Figure 4.3 presents the Level 1 Data Flow Diagram of the proposed TBML detection framework by decomposing the primary system into multiple functional processes. The architecture consists of modules including Data Collection, Data Preprocessing, Heterogeneous Graph Construction, Federated Learning Engine, HGNN Fraud Detection, and Alert and Monitoring Dashboard. The Data Collection module acquires raw transactional records and customer information from financial institutions, which are subsequently cleaned and transformed within the preprocessing stage to generate structured financial features. These processed features are utilized by the graph construction engine to build heterogeneous graph representations stored within the graph database. The federated learning engine performs distributed collaborative model training using encrypted local updates received from federated client nodes while maintaining institutional data privacy. The trained HGNN model is then employed by the fraud detection engine to identify suspicious transactional patterns, calculate fraud scores, and generate anomaly alerts. Finally, the monitoring dashboard enables AML analysts and compliance teams to perform investigations, visualize transaction graphs, and access risk assessment reports

for real-time anti-money laundering operations.

2.4.3 Data Flow Diagram Level 2

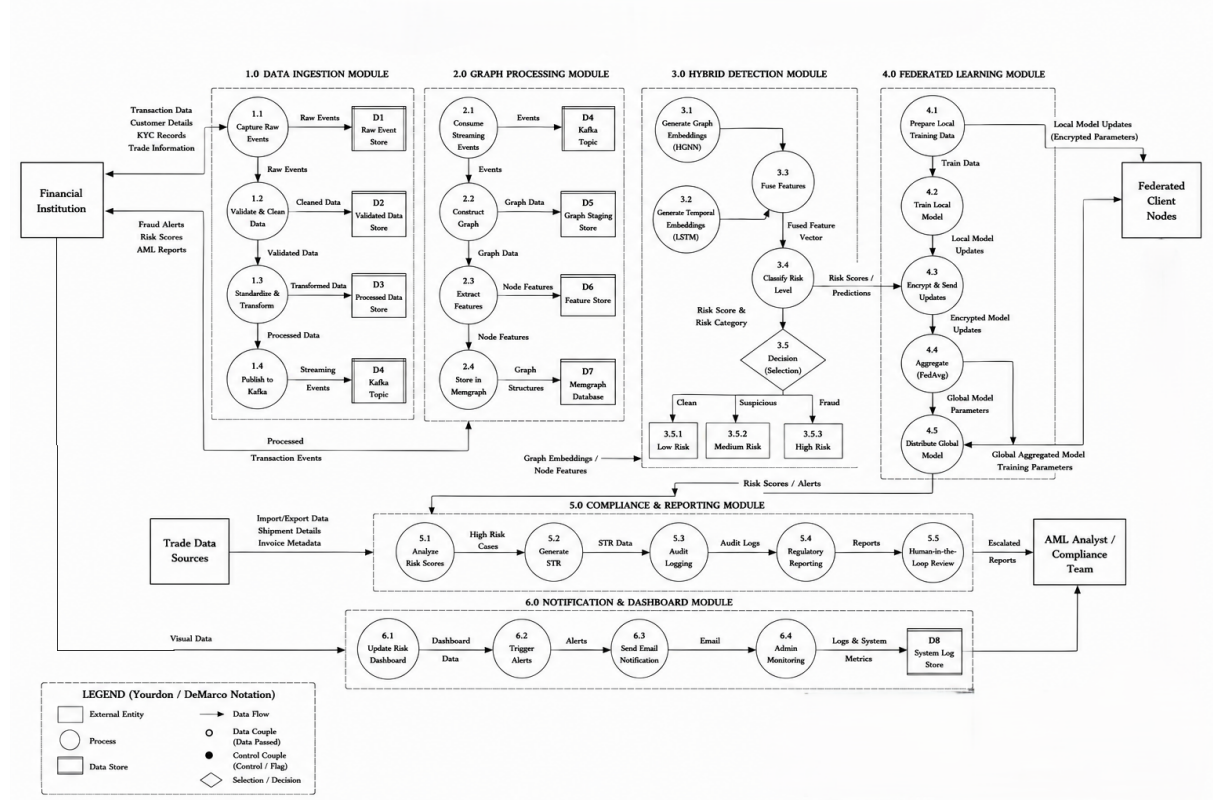


Figure 2.4: Level 2 Data Flow Diagram

Figure 4.4 illustrates the Level 2 Data Flow Diagram representing the internal decomposition of the HGNN Fraud Detection module within the proposed system. The process begins with Data Integration and Validation, where trade invoice records, shipment details, and transactional metadata are verified and filtered for inconsistencies. The validated information is then processed through Entity Resolution and Feature Extraction to identify relational attributes and generate contextual financial features required for graph-based learning. Subsequently, the Heterogeneous Graph Generation process constructs interconnected financial transaction graphs representing entities, accounts, invoices, and trade relationships. The HGNN Inference and Risk Scoring module applies graph neural network-based learning to calculate anomaly scores and identify suspicious transactional behavior. The resulting outputs are further processed through Anomaly Detection and Classification, followed by Alert Generation and Prioritization to rank suspicious activities according to calculated risk severity. Finally, the Result Aggregation

and Reporting module generates consolidated fraud summaries, suspicious activity alerts, and analytical reports that are forwarded to the Alert and Monitoring Dashboard for investigation and compliance analysis. The Level 2 DFD provides a detailed representation of the internal analytical workflow employed by the proposed federated graph intelligence framework for real-time TBML detection.

2.5 Summary

This chapter presented the high-level design of the proposed Federated Heterogeneous Graph Neural Network (HGNN)-Based Trade-Based Money Laundering (TBML) Detection Framework. The design considerations focused on scalability, privacy preservation, modularity, real-time transaction monitoring, and secure institutional collaboration within distributed financial environments. The chapter also discussed the development methodology, architectural strategies, programming technologies, and system integration mechanisms utilized for implementing graph-based fraud analysis, federated learning, and compliance monitoring services.

The overall system architecture demonstrated the integration of transaction processing modules, heterogeneous graph construction, hybrid graph neural network models, federated collaborative learning, and compliance reporting environments for intelligent TBML detection. Additionally, the chapter presented the Level 0, Level 1, and Level 2 Data Flow Diagrams to illustrate the movement of financial data, internal analytical workflows, graph-based fraud detection processes, and alert generation mechanisms within the proposed framework. These architectural components collectively establish a scalable, privacy-aware, and intelligent monitoring system for real-time anti-money laundering analysis.



Appendix A

Plagiarism report

APPENDIX A

PLAGIARISM REPORT

Attach the screenshot of plagiarism report front page.





Appendix B

Publication details

APPENDIX B

PUBLICATION DETAILS

Attach the screenshot of either the Acknowledgment of the publication / Proof of Acceptance / Proof of Registration.





Appendix C

Code

APPENDIX C

CODE

C.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```